Ofqual QAN	**600/6407/9**
Quartz ID	**1047**
Qualification	**Level 3 Diploma in Computing**

Assessment	**Mathematical Skills for Computing**
Assessment Type	**Examination**
Date	**Wednesday 28th May 2025**
Time	**9.00am – 12.00pm**

Examination Paper

Answer ALL questions.

Clearly cross out surplus answers.

Time: 3 hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination.

Answer ALL questions

Marks

Question 1

Select the option which is equal to $a^{4/5}$: **1**

a. a^1 b. a^{-1}

c. $\sqrt[4]{a^5}$ d. $\sqrt[5]{a^4}$

Question 2

Expand and simplify the expression: **1**

$$3x(7x - 4y)$$

a. $21x^2 + 12xy$ b. $10x - 4y$

c. $21x^2 - 12xy$ d. $10x - 3x4y$

Question 3

Simplify the expression: **1**

$$(b - 3)(b - 5)$$

a. $b^2 - 8b + 15$ b. $b^2 + 8b + 15$

c. $b^2 - 8b - 15$ d. $b^2 + 8b - 15$

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Marks

Question 4

Factorise the expression: **1**

$$16 - 9c^2$$

a. $(4 - 3c)(4 - 3c)$

b. $(4 + 3c)(4 + 3c)$

c. $(4 + 3c)(4 - 3c)$

d. $(4 - 9c)(4 - c)$

Question 5

Rearrange the following formula to make d the subject: **1**

$$7d^2 - e = 5$$

a. $d = \pm\sqrt{\dfrac{5 - e}{7}}$

b. $d = \left(\dfrac{5 + e}{7}\right)^2$

c. $d = \pm\sqrt{\dfrac{5+e}{7}}$

d. $d = \sqrt{\dfrac{7}{5 + e}}$

Question 6

Find the value of x when **1**

$$2^{x+3} = 256$$

a. $x = 3$

b. $x = 128$

c. $x = 8$

d. $x = 5$

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Marks

Question 7

The sum of three consecutive numbers is 366. Select the equation which represents this in terms of x (where x represents the first number). **1**

a. $3x + 1 = 366$ b. $3x - 1 = 366$

c. $x + 3 = 366$ d. $3x + 3 = 366$

Question 8

Select the y-intercept of the line $y = 3x - \frac{1}{2}$: **1**

a. x b. $\frac{1}{2}$

c. 3 d. $-\frac{1}{2}$

Question 9

Select the option which is an example of a null set. **1**

a. The set of prime numbers less than 53
b. The set of factors of 15 that are even numbers
c. The set of numbers divisible by 21
d. The set of multiples of 11 that are even numbers

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Marks

Question 10

If A = $\{3, 7, 8, 12\}$ and $B = \{9, 15\}$ select the correct option. **1**

a. $A \subset B$ b. $B \subset A$
c. A and B are disjoint sets. d. A and B are infinite sets.

Question 11

If C = $\{15, 25, 28, 91, 101, 124, 171, 218, 220\}$ find $\mathrm{n(C)}$. **1**

a. 9 b. 205
c. 220 d. $\{15, 220\}$

Question 12

If $D = \{5, 9, 17, 21, 28, 30, 34\}$ and $E = \{5, 17, 30\}$, find $D - E$. **1**

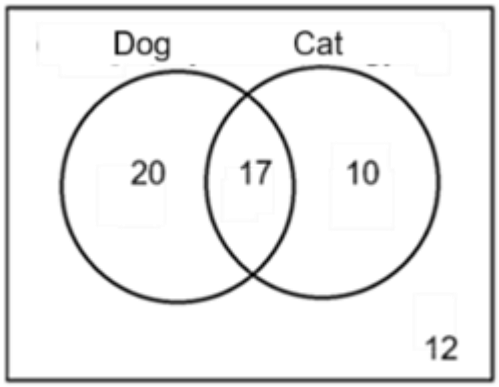
a. $\{9, 21, 28, 34\}$ b. $\{4\}$
c. $\{5, 34\}$ d. $\{31\}$

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Marks

Question 13

A group of customers in a pet shop are surveyed and asked if they own any pets. The responses are displayed in the Venn diagram below: **1**

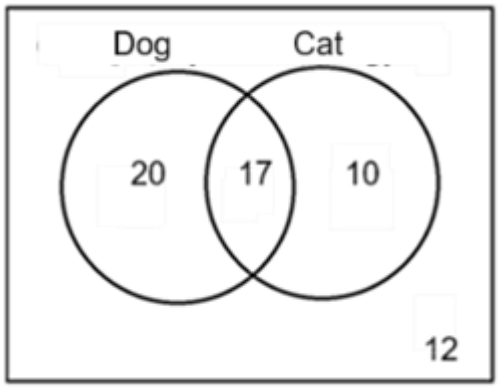


Find the number of customers who do **not** own a cat.

a. 49
b. 32
c. 37
d. 12

Question 14

Using the same Venn diagram as Q13: **1**



Find the number of customers who own a dog.

a. 37
b. 20
c. 17
d. 49

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Marks

Question 15

Select the option that gives the order of the following matrix: **1**

$$\begin{pmatrix} 3 & 2 \\ 7 & 1 \\ 8 & 4 \end{pmatrix}$$

a. 3
b. 6
c. 3 × 2
d. 2 × 3

Question 16

TWO (2) matrices are given as: **1**

$$F = \begin{pmatrix} -2 & 8 \\ 5 & 2 \\ -3 & 4 \end{pmatrix} \text{ and } G = \begin{pmatrix} -1 & 2 \\ -6 & 1 \\ 8 & -4 \end{pmatrix}$$

Calculate F+G.

a. $\begin{pmatrix} 0 & 6 & -7 \\ 7 & -4 & 12 \end{pmatrix}$

b. $\begin{pmatrix} -3 & -1 & 5 \\ 10 & 3 & 0 \end{pmatrix}$

c. $\begin{pmatrix} 0 & 7 \\ 6 & -4 \\ -7 & 12 \end{pmatrix}$

d. $\begin{pmatrix} -3 & 10 \\ -1 & 3 \\ 5 & 0 \end{pmatrix}$

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Marks

Question 17

TWO (2) matrices are given as: **1**

$$J = (-4 \quad -2 \quad 2) \text{ and } K = \begin{pmatrix} -2 \\ -1 \\ 3 \end{pmatrix}$$

Calculate JK.

a. $\begin{pmatrix} 8 \\ 2 \\ 6 \end{pmatrix}$ b. 16

c. $(8 \quad 2 \quad 6)$ d. -16

Question 18

Find the transpose of the following matrix: **1**

$$\begin{pmatrix} 3 & 1 & -2 \\ 1 & 2 & 1 \end{pmatrix}$$

a. $\begin{pmatrix} 3 & 1 \\ 1 & 2 \\ -2 & 1 \end{pmatrix}$ b. $\begin{pmatrix} -3 & -1 & 2 \\ -1 & -2 & -1 \end{pmatrix}$

c. $\begin{pmatrix} 1 & 2 & 1 \\ 3 & 1 & -2 \end{pmatrix}$ d. $\begin{pmatrix} 1 & 3 \\ 2 & 1 \\ 1 & -2 \end{pmatrix}$

Marks

Question 19

Find the determinant of M if $M = \begin{pmatrix} 2 & 1 \\ -3 & 8 \end{pmatrix}$ **1**

a. -14 b. 14
c. 13 d. 19

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Question 20

A shop records the number of online sales received each day. Select the option which describes the data collected. **1**

a. Secondary
b. Qualitative
c. Discrete
d. Continuous

Question 21

A group of people were surveyed to find out their preferred food. The results are displayed in the pie chart below: **1**



Select the most popular fruit.

a. Apple
b. Banana
c. Orange
d. Pineapple

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Marks

Question 22

Using the same pie chart as Question 21: **1**



Select the proportion of people surveyed whose favourite fruit is apple.

a. 0.2
b. 0.72
c. 0.25
d. 72

Question 23

Using the same pie chart as Question 21: **1**



The total number of people who took part in the survey is 60. Select the number of people whose favourite fruit is orange.

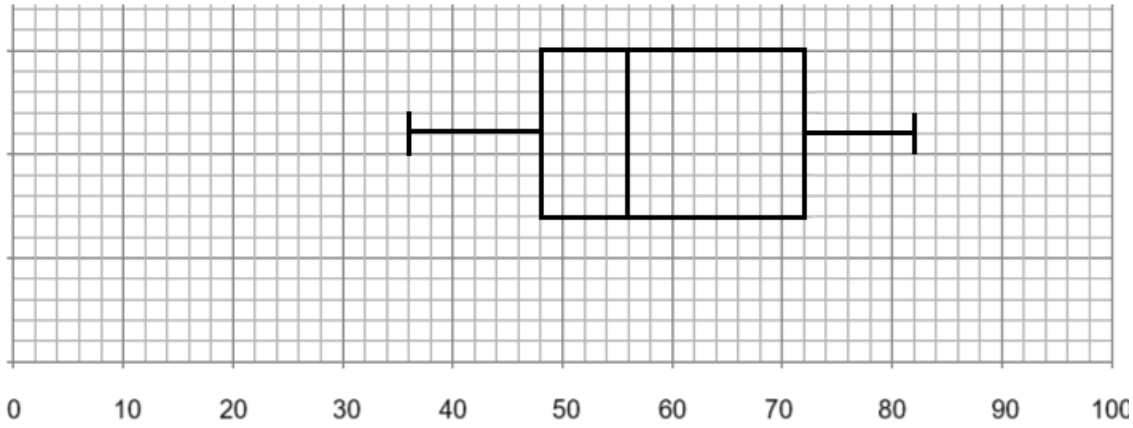
a. 45
b. 15
c. 60
d. 30

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Marks

Question 24

A survey collects data on how long it takes a sample of people to travel to work. The results are recorded as a boxplot below: **1**

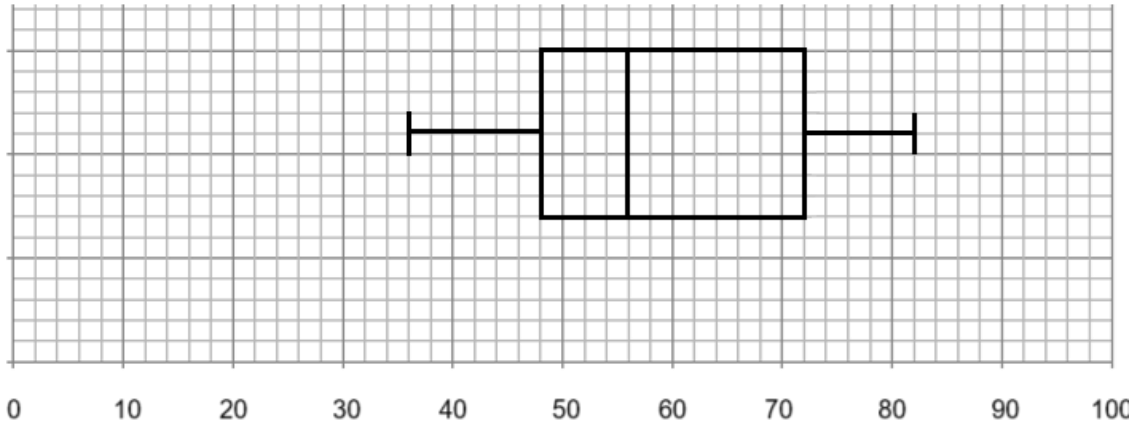


Select the option which gives the range of the data.

a. 56 minutes
b. 24 minutes
c. 46 minutes
d. 72 minutes

Question 25

Using the same box plot as in Question 24 and reproduced below: **1**



Select the option which gives the median of the data.

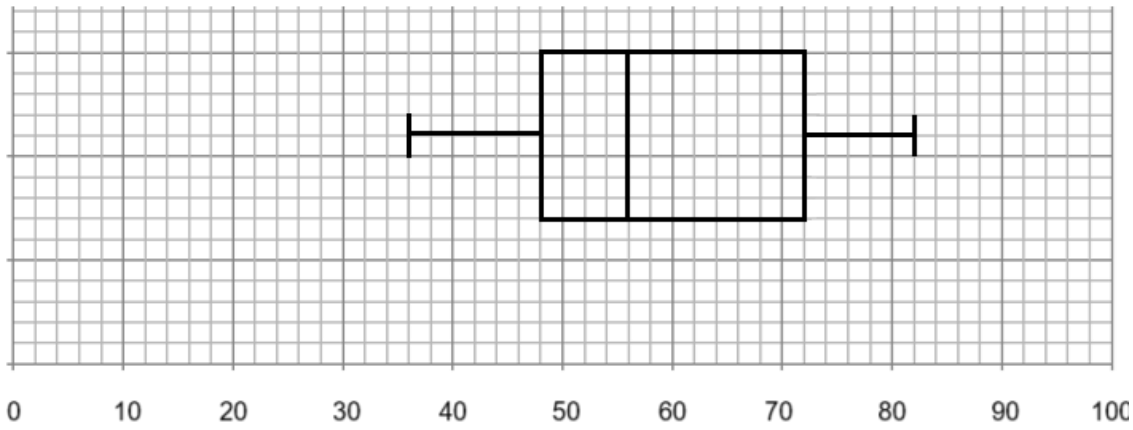
a. 56 minutes
b. 48 minutes
c. 46 minutes
d. 24 minutes

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Marks

Question 26

Using the same box plot as in Question 24 and reproduced below: **1**



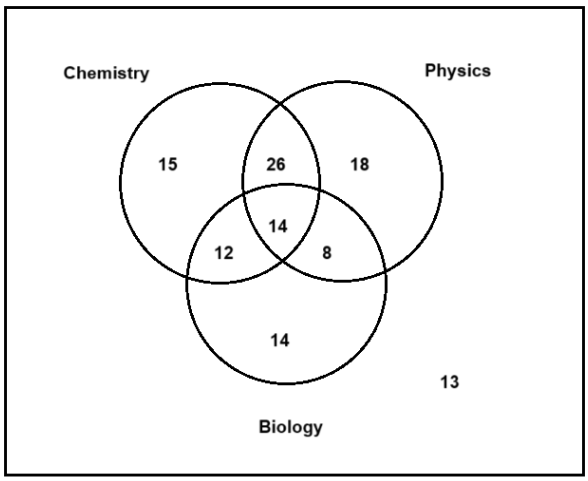
Select the option which gives the interquartile range of the data.

a. 46 minutes
b. 24 minutes
c. 56 minutes
d. 82 minutes

Marks

Question 27

ONE HUNDRED AND TWENTY (120) students were asked which science they studied; chemistry, physics or biology. Their responses are recorded in the Venn diagram below: **1**



Find the probability that a student, selected at random, studied physics.

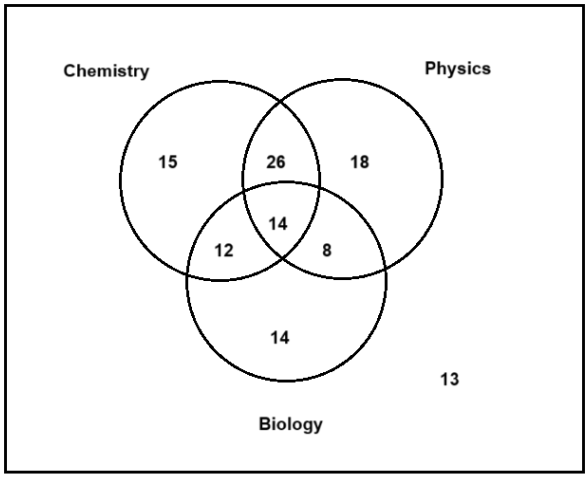
a. $\frac{18}{120}$
b. $\frac{11}{20}$
c. $\frac{18}{66}$
d. 66

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Marks

Question 28

Using the same Venn diagram as Question 27 and reproduced below: **1**



Find the probability that a student, selected at random, studied both chemistry and biology.

a. 26

b. $\frac{29}{120}$

c. $\frac{1}{10}$

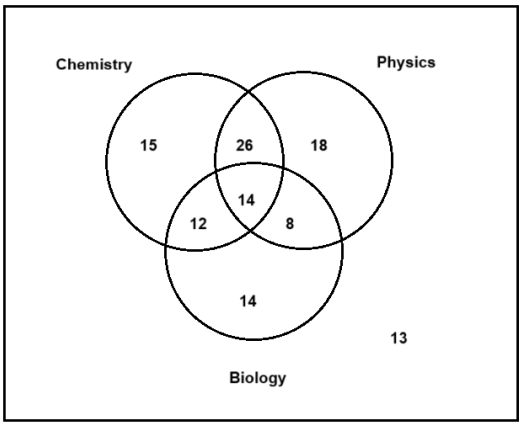
d. $\frac{13}{60}$

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Marks

Question 29

Using the same Venn diagram as Question 27 and reproduced below: **1**



Find the probability that a student, selected at random, studied physics but not biology.

a. $\frac{29}{60}$ b. $\frac{11}{30}$

c. $\frac{3}{20}$ d. 44

Question 30

A fair, EIGHT (8) -sided spinner, numbered ONE (1) to EIGHT (8) is rolled. **1**
Find the theoretical probability of the spinner landing on a TWO (2).

a. 2 b. $\frac{1}{8}$

c. $\frac{2}{8}$ d. $\frac{8}{1}$

Total 30 marks

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Question 31

a) Simplify the following:

i) $\frac{a^5}{a^2} \times (a^4)^2$ **1**

ii) $b^0 + c^0 - 3d^0$ **1**

iii) $c^{-5} \times c^{-2} \times c^8$ **1**

b) Completely factorise: **2**

$$x^2 + 4x - 21$$

c) Solve the following quadratic equation by using the Quadratic Formula: **2**

$$4x^2 - 17x + 12 = 0$$

Leave your answer in surd form.

d) Solve the following simultaneous equations algebraically and find the value of x and y: **3**

$$10x + y = 8 \text{ and } y = 12 - 18x$$

Total 10 Marks

Question 32

a) State the gradient of the line $y = -2x + 5$ **1**

b) Consider the pair of simultaneous equations:

$$y = 2x + 3 \quad \text{and } y = -\frac{2}{3}x + 1$$

i) Construct a table of integer values for each equation for the range: **4**

$$-3 \leq x \leq 1$$

ii) Draw accurate graphs for both equations on ONE (1) set of axes. Use the graph paper provided. **4**

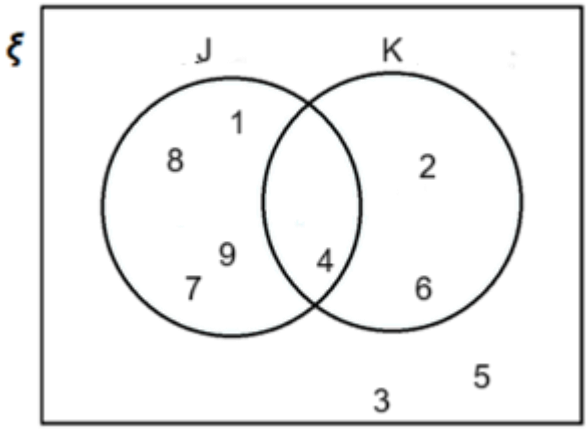
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iii) Use the graphs constructed in (ii) to find the solution of the pair of simultaneous equations. **1**

Total 10 Marks

Question 33

a)



Use the Venn diagram above to state which numbers are in the set:

i) J' **1**

ii) $J \cap K$ **1**

iii) $J' \cap K$ **1**

iv) $J' \cup K'$ **1**

b) $M = \{1, 11, 17, 25\}$ and $N = \{1,7, 9.11,12, 17,20, 21, 25\}$ **2**
State whether $M \subset N$. Explain your answer.

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c) TWENTY- FIVE (25) students were asked about which type of fruit juice they like. **4**

- TEN (10) only like orange juice
- SEVEN (7) only like apple juice
- FIVE (5) like both orange juice and apple juice.

Draw a Venn diagram to represent this data.

Total 10 Marks

Question 34

a) TWO (2) matrices are given as:

$$P = \begin{pmatrix} 2 & 4 \\ -3 & 1 \end{pmatrix} \text{ and } Q = \begin{pmatrix} 1 & -1 \\ -2 & 2 \end{pmatrix}$$

i) Calculate $P - Q$. **1**

ii) Calculate $-5Q$ **1**

iii) Find P^2 **2**

b) TWO (2) matrices are given as: **3**

$$M = \begin{pmatrix} 3 & -1 \\ 2 & -4 \end{pmatrix} \text{ and } N = \begin{pmatrix} 2 & 1 \\ -1 & -2 \end{pmatrix}$$

Find the product MN and then apply it to the coordinate vector $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$.

c) Describe fully the single transformation represented by $\begin{pmatrix} 7 & 0 \\ 0 & 7 \end{pmatrix}$. **3**

Total 10 Marks

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Question 35

A café records the average daily temperature and number of hot drinks sold over a 14-day period. The table below shows these data:

Day	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Average daily temperature (°C)	15	14	19	20	21	23	26	17	14	16	24	23	21	25
Number of hot drinks sold.	50	44	28	32	25	20	10	34	51	43	11	12	20	8

a) Draw a scatter graph for this data using the graph paper provided. You should plot the average daily temperature on the x-axis and the number of hot drinks sold on the y-axis. **4**

b) Draw a line of best fit on the scatter graph. **1**

c) Describe the relationship between the average daily temperature (°C) and the number of hot drinks sold. State the name given to this type of correlation. **2**

d) Use the line of best fit to estimate the number of hot drinks sold when the average daily temperature is 22 °C. State the name given to this type of estimate. **2**

e) On day 15 the average temperature is 24 °C and the café sells 50 hot drinks. State the name given to this data point. **1**

Total 10 Marks

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Question 36

This question uses the data about the number of hot drinks sold over a 15 day period from Q35. The data are recorded below.

Day	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Number of hot drinks sold.	50	44	28	32	25	20	10	34	51	43	11	12	20	8	50

a) State whether the data collected about the number of hot drinks sold is quantitative or qualitative. Briefly explain your answer. **2**

b) Calculate the mean number of hot drinks sold. Give your answer to the nearest whole number. Show your working. **2**

c) Calculate the variance of the number of hot drinks sold. Give your answer to the nearest whole number. Show your working. **3**

d) Calculate the standard deviation of the number of hot drinks sold. Show your working. **1**

e) Calculate the coefficient of variation of the data. Express your answer as a percentage rounded to 3 significant figures. **2**

Total 10 Marks

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Question 37

A box contains two red pens and seven blue pens. Two pens are selected at random without replacement.

a) Draw a *probability tree diagram* to show this information and all the possible outcomes. You should state all possible outcomes **and** calculate the probability of each outcome. **8**

b) Use your tree diagram to find out the *probability* that both pens selected are the **same** colour. Do not round your answer. **2**

Total 10 Marks

End of Paper

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Mathematical Skills for Computing

Formulae sheet

Table of Content

1. The Quadratic Formula and Discriminants
2. Matrices
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1. The Quadratic Formula and Discriminants

Written in standard form, $ax^2 + bx + c = 0$, any quadratic equation can be solved using the **quadratic formula**:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

where a, b, and c are real numbers and $a \neq 0$

The discriminant is defined as b^2-4ac.

- If $b^2 - 4ac = 0$, there is one rational solution
- If $b^2 - 4ac > 0$ and a perfect square, there are two rational solutions
- If $b^2 - 4ac > 0$ and not a perfect square, there are two irrational solutions
- If $b^2 - 4ac < 0$, there are two complex solutions

2. Matrices

Addition and subtraction

If $\boldsymbol{A} = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} \end{bmatrix}$ and $\boldsymbol{B} = \begin{bmatrix} b_{1,1} & b_{1,2} & \cdots & b_{1.n} \\ b_{2,1} & b_{2,2} & \cdots & b_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m,1} & b_{m,2} & \cdots & b_{m,n} \end{bmatrix}$, then

$$\boldsymbol{A} + \boldsymbol{B} = \begin{bmatrix} a_{1,1} + b_{1,1} & a_{1,2} + b_{1,2} & \cdots & a_{1,n} + b_{1,n} \\ a_{2,1} + b_{2,1} & a_{2,2} + b_{2,2} & \cdots & a_{2,1} + b_{2,1} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} + b_{m,1} & a_{m,2} + b_{m,2} & \cdots & a_{m,n} + b_{m,n} \end{bmatrix}$$ and

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$$\boldsymbol{A}-\boldsymbol{B}=\begin{bmatrix} a_{1,1}-b_{1,1} & a_{1,2}-b_{1,2} & \cdots & a_{1,n}-b_{1,n} \\ a_{2,1}-b_{2,1} & a_{2,2}-b_{2,2} & \cdots & a_{2,1}-b_{2,1} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1}-b_{m,1} & a_{m,2}-b_{m,2} & \cdots & a_{m,n}-b_{m,n} \end{bmatrix}$$

Scalar multiplication

If a matrix, $\boldsymbol{A}$, is multiplied (or scaled) by a constant k, then

$$k\boldsymbol{A}=\begin{bmatrix} ka_{1,1} & ka_{1,2} & \cdots & ka_{1,n} \\ ka_{2,1} & ka_{2,2} & \cdots & ka_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ ka_{m,1} & ka_{m,2} & \cdots & ka_{m,n} \end{bmatrix}$$

The matrix product

If $\boldsymbol{A}=\begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} \end{bmatrix}$ and $\boldsymbol{C}=\begin{bmatrix} c_{1,1} & c_{1,2} & \cdots & c_{1,p} \\ c_{2,1} & c_{2,2} & \cdots & c_{2,p} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n,1} & c_{n,2} & \cdots & c_{n,p} \end{bmatrix}$ then

the matrix product, $\boldsymbol{AC}$

$$\boldsymbol{AC}=\begin{bmatrix} a_{1,1}b_{1,1}+\cdots+a_{1,n}b_{n,1} & a_{1,1}b_{1,2}+\cdots+a_{1,n}b_{n,2} & \cdots & a_{1,1}b_{1,p}+\cdots+a_{1,n}b_{n,p} \\ a_{2,1}b_{1,1}+\cdots+a_{2,n}b_{n,1} & a_{2,1}b_{1,2}+\cdots+a_{2,n}b_{n,2} & \cdots & a_{2,1}b_{1,p}+\cdots+a_{2,n}b_{n,p} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1}b_{1,1}+\cdots+a_{m,n}b_{n,1} & a_{m,1}b_{1,2}+\cdots+a_{m,n}b_{n,2} & \cdots & a_{m,1}b_{1,p}+\cdots+a_{m,n}b_{n,p} \end{bmatrix}$$

Determinant

The determinant of a square matrix is calculated from its elements. The determinant of a 2×2 matrix $\boldsymbol{D}=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is given by $\det \boldsymbol{A}=ad-bc$

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3. Statistics

Mean

Ungrouped data:

$$\bar{x} = \frac{\sum x}{n}$$

Ungrouped frequency table:

$$\bar{x} = \frac{\sum fx}{n} = \frac{\sum fx}{\sum f}$$

Grouped frequency table:

$$\bar{x} = \frac{\sum fm}{n} = \frac{\sum fm}{\sum f}$$

Mode

Grouped data

Estimated mode = $L + \frac{(F_m - F_b) \times c}{(2F_m - F_b) - F_n}$

Median

Grouped data

Estimated median = $L_{med} + W_{med} \times \frac{\frac{n}{2} - C_{med}}{F_{med}}$

Quartiles

Ungrouped data:

$$LQ = \frac{1}{4}(n+1)$$

$$UQ = \frac{3}{4}(n+1)$$

Grouped data

Quartile = $L + \frac{(Q-P) \times I}{F}$

Mean deviation

Ungrouped data

Mean deviation $= \frac{\sum |x - \bar{x}|}{n}$

Grouped data

Mean deviation $= \frac{\sum f|x - \bar{x}|}{\sum f}$

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Variance

Ungrouped data:

$$s^2 = \frac{\sum(x - \bar{x})^2}{n} = \frac{\sum x^2}{n} - \bar{x}^2$$

Grouped data:

$$s^2 = \frac{\sum f(x - \bar{x})^2}{\Sigma f} = \frac{\sum f x^2}{\Sigma f} - \left[\frac{\sum f x}{\Sigma f}\right]^2$$

Standard deviation

Ungrouped data:

$$s = \sqrt{\frac{\sum(x - \bar{x})^2}{n}}$$

Grouped data:

$$s = \sqrt{\frac{\sum f x^2}{\Sigma f} - \left[\frac{\sum f x}{\Sigma f}\right]^2}$$

Coefficient of Variation

$$CV = \frac{s}{\bar{x}}$$

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